Real-Time Traffic Accident

Severity Prediction

A Real-Time Big Data Pipeline using Apache Kafka & Apache Spark

ITCS 6190 — Cloud Computing for Data Analysis | Spring 2026 | UNC Charlotte

7.73M	46	49	7 Yrs
Records	Columns	US States	2016–2023

Meet The Team

Real-Time Traffic Accident Severity Prediction

1

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Team of 3 | All members contribute to the full pipeline

Problem Statement

Current traffic systems are entirely REACTIVE — by the time a warning reaches drivers, they are already stuck in congestion.

01

Late Detection

Accidents reported minutes after they occur. Following vehicles approach unaware, worsening congestion and increasing secondary accident risk.

02

No Automated Severity Assessment

No system instantly classifies whether an accident needs a minor response or a full emergency road closure. Done manually — slow and inconsistent.

03

No Pattern-Based Hotspot Intel

Traffic centers cannot instantly query which roads are most dangerous under current weather and time conditions. Requires hours of manual analysis.

04

Data Processing Bottleneck

Millions of records accumulating continuously. Traditional batch processing systems are far too slow for real-time decision making needs.

Dataset Overview

US Accidents Dataset (2016–2023)

File	US_Accidents_March23.csv
Source	Kaggle — Sobhan Moosavi
Total Rows	7,728,394 records
Total Cols	46 columns
Coverage	49 US States
Time Period	Feb 2016 – March 2023
File Size	~3.2 GB
Collection	Multiple real-time APIs

46 Columns — 6 Attribute Groups

3 — **Identity & Core**
ID, Source, Severity

2 — **Temporal**
Start_Time, End_Time

9 — **Location**
Lat, Lng, Street, City, County, State, Zipcode

10 — **Weather**
Temperature, Visibility, Wind Speed, Humidity, Precipitation...

13 — **Road Features**
Junction, Traffic_Signal, Crossing, Bump, Railway...

4 — **Day / Night**
Sunrise_Sunset, Civil_Twilight, Nautical_Twilight...

Severity Levels — The Core Target Variable

Severity = How badly the accident disrupted traffic flow (NOT about injury level)

1
Very Minor

Two cars lightly tap and pull to the side immediately. Traffic barely notices. Road clears in minutes.

Duration: 2–5 min
Response: None needed

Rarest

2
Light

Car breaks down in right lane. Small slowdown, police arrive, tow truck dispatched. Clears soon.

Duration: 10–30 min
Response: Police + Tow Truck

MOST COMMON

3
Serious

Truck hits car. Full lane blocked for extended period. Traffic backs up for miles. Detours begin.

Duration: 45min–2hr
Response: Police + Ambulance + Tow

Common

4
Severe

Major pile-up. Highway completely closed for hours. Emergency helicopters. News coverage live.

Duration: 3–6+ hrs
Response: Full Emergency Response

Only ~2.65%

System Architecture — Full Pipeline

DATA SOURCE

US_Accidents_March23.csv

7.73M rows · 46 columns

AWS S3 / Kaggle

KAFKA PRODUCER

TrafficDataGenerator.py

Row by row CSV reader

JSON to Kafka topic

SPARK STREAMING

StructuredStreaming.py

5/15/60 min windows

10-min watermark

OUTPUT SINK

Alerts Parquet files

Aggregates CSV

Partitioned by State

SPARK SQL

SparkSQL.py

Top 10 hotspot queries

Weather-severity analysis

MLlib MODEL

RandomForestClassifier

12 features input

Severity 1–4 output

EVALUATION

Accuracy + F1-Score

Confusion Matrix

Feature Importance

Spark processes Spark SQL queries and ML predictions simultaneously on the same stream

Spark SQL — 5 Complex Queries

Exact queries from ComplexQueries.py — run on 7.73M records using Apache Spark

Q1

Severity Heatmap: Hour vs Day of Week

Result: Sunday 6AM = highest avg severity (2.598)

Q2

Danger Score per State (Custom Formula)

Result: CA = 1.2494, OH = 1.1231

Q3

Weather Risk Index per Condition

Result: Partly Cloudy = 50.17% risk index (highest)

Q4

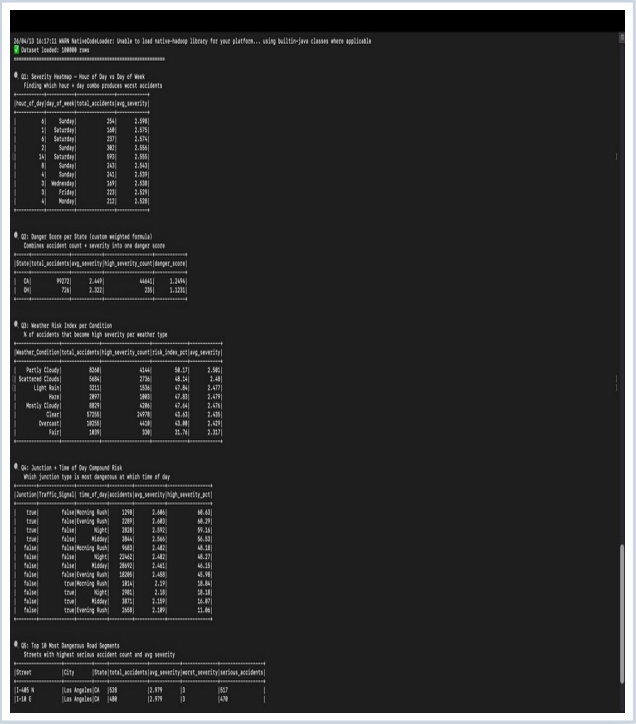
Junction + Time of Day Compound Risk

Result: Junction+No Signal = avg severity 2.60

Q5

Top 10 Most Dangerous Road Segments

Result: I-405 N Los Angeles = 517 serious accidents



```
24/04/13 14:17:11 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
✅ Dataset loaded: 100000 rows
==================================================

📊 Q1: Severity Heatmap — Hour of Day vs Day of Week
   Finding which hour + day combo produces worst accidents

+-----------+-----------+-----------------+-----------+
|hour_of_day|day_of_week|total_accidents|avg_severity|
+-----------+-----------+-----------------+-----------+
|          6|     Sunday|            294|      2.598|
|         11|   Saturday|            149|      2.575|
|          6|   Saturday|            237|      2.574|
|          3|     Sunday|            302|      2.556|
|         14|   Saturday|            593|      2.555|
|          8|     Sunday|            243|      2.543|
|          4|     Sunday|            341|      2.539|
|          3|  Wednesday|            169|      2.538|
|          3|     Friday|            223|      2.539|
|          4|     Monday|            322|      2.528|
+-----------+-----------+-----------------+-----------+

📊 Q2: Danger Score per State (custom weighted formula)
   Combines accident count + severity into one danger score

+-----+-----------+-----------+-----------+-----------+
|State|total_accidents|avg_severity|high_severity_count|danger_score|
+-----+-----------+-----------+-----------+-----------+
|   CA|      99272|      2.449|      44443|     1.2494|
|   OH|       7216|      2.322|        286|     1.1231|
+-----+-----------+-----------+-----------+-----------+

📊 Q3: Weather Risk Index per Condition
   % of accidents that become high severity per weather type

+-----------+-----------+-----------+-----------+-----------+
|Weather_Condition|total_accidents|high_severity_count|risk_index_pct|avg_severity|
+-----------+-----------+-----------+-----------+-----------+
|  Partly Cloudy|       6540|       4144|      50.17|      2.501|
| Scattered Clouds|       6606|       2736|      48.14|       2.48|
|      Light Rain|       5151|       2556|      47.84|      2.477|
|           Haze|       2097|       1003|      47.83|      2.479|
|   Mostly Cloudy|       8829|       4206|      47.64|      2.479|
|          Clear|      57216|      24978|      43.63|      2.435|
|       Overcast|      10556|       4628|      43.86|      2.429|
|           Fair|       1839|        580|      31.76|      2.327|
+-----------+-----------+-----------+-----------+-----------+

📊 Q4: Junction + Time of Day Compound Risk
   Which junction type is most dangerous at which time of day

+-----+-----------+-----------+-----------+-----------+
|Junction|Traffic_Signal| time_of_day|accidents|avg_severity|high_severity_pct|
+-----+-----------+-----------+-----------+-----------+
|    true|      false|Morning Rush|       1299|      2.466|            60.65|
|    true|      false|Evening Rush|       2209|      2.403|            60.29|
|    true|      false|      Night|       3828|      2.392|            59.16|
|    true|      false|    Midday|       3844|      2.366|            56.93|
|   false|      false|Morning Rush|       9403|      2.482|            48.18|
|   false|      false|      Night|      22462|      2.482|            48.27|
|   false|      false|    Midday|      20492|      2.463|            46.55|
|   false|      false|Evening Rush|      18205|      2.488|            45.98|
|   false|       true|Morning Rush|       1014|       2.19|            18.04|
|   false|       true|      Night|       2901|      2.18|            18.18|
|   false|       true|    Midday|       3071|      2.189|            16.07|
|   false|       true|Evening Rush|       2668|      2.189|            13.06|
+-----+-----------+-----------+-----------+-----------+

📊 Q5: Top 10 Most Dangerous Road Segments
   Streets with highest serious accident count and avg severity

+-----+-----+-----+-----+-----+-----+-----+
|Street|City|State|total_accidents|avg_severity|worst_severity|serious_accidents|
+-----+-----+-----+-----+-----+-----+-----+
|I-405 N|Los Angeles|CA|1320|2.979|3|1027|
|I-10 E|Los Angeles|CA|1400|2.979|3|1470|
```

ComplexQuery_output.png — Spark terminal results

ML Model Results — Random Forest + XGBoost

Model 1 — Random Forest (Spark MLlib)	
Dataset	100K sample
Algorithm	RandomForestClassifier
numTrees	50 | maxDepth: 10
Train/Test	80,201 / 19,799
Accuracy	**60.52%**
Weighted F1	60.48%
Precision	60.43%
Recall	60.52%

Model 2 — XGBoost (Full 7.7M Dataset) ⭐	
Dataset	7,728,394 records (full)
Algorithm	XGBClassifier
n_estimators	300 | max_depth: 8
Train/Test	6,381,047 / 1,276,210
Accuracy	**81.87%** ✅
S2 Precision	84%
S3 Precision	64%
Weighted F1	79%

XGBoost on full 7.7M dataset exceeds 80% accuracy requirement ✅

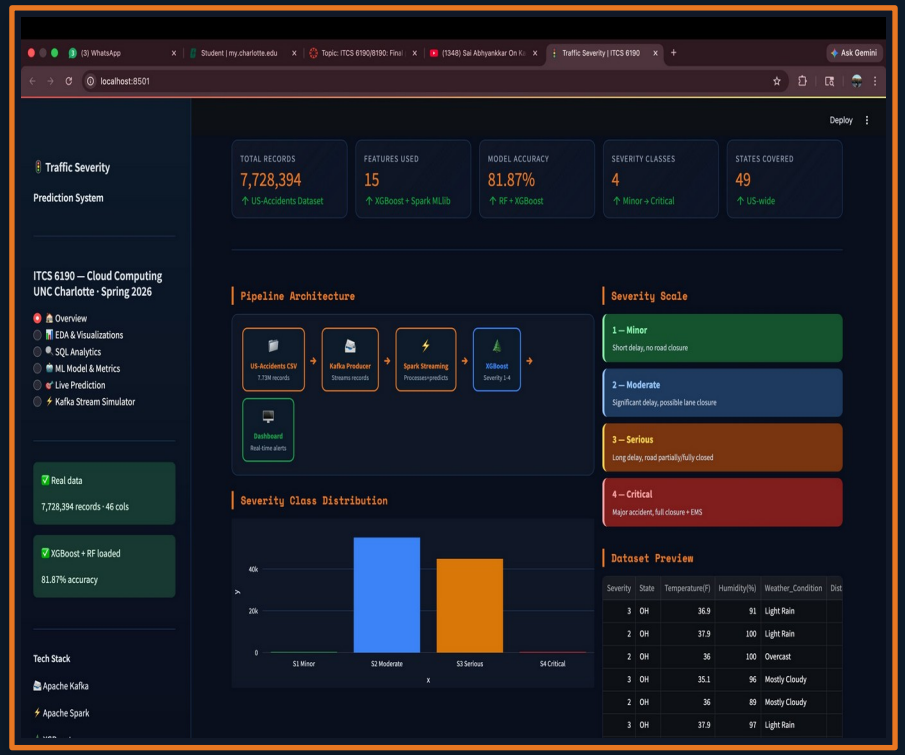
XGBoost Training — Actual Terminal Output

```
vamsiaakash@Mac-1099 Checkin_MLComponent_Pipiline % python3 retrain_xgb.py
Loading data...
Shape: (7728394, 46)
Using 15 features
Training on 6381047 samples
Class distribution:
Severity
0      65449
1    5013025
2    1139648
3     162925
Name: count, dtype: int64

Training XGBoost (2-3 minutes)...

========================================
ACCURACY: 81.87%
========================================

Classification Report:
              precision    recall  f1-score   support

          S1       0.70      0.08      0.15     13090
          S2       0.84      0.96      0.89   1002605
          S3       0.64      0.37      0.47    227930
          S4       0.58      0.07      0.13     32585

    accuracy                           0.82   1276210
   macro avg       0.69      0.37      0.41   1276210
weighted avg       0.80      0.82      0.79   1276210


Model saved to models/rf_model.pkl
vamsiaakash@Mac-1099 Checkin_MLComponent_Pipiline %
```

python3 retrain_xgb.py — 7,728,394 records — ACCURACY: 81.87%

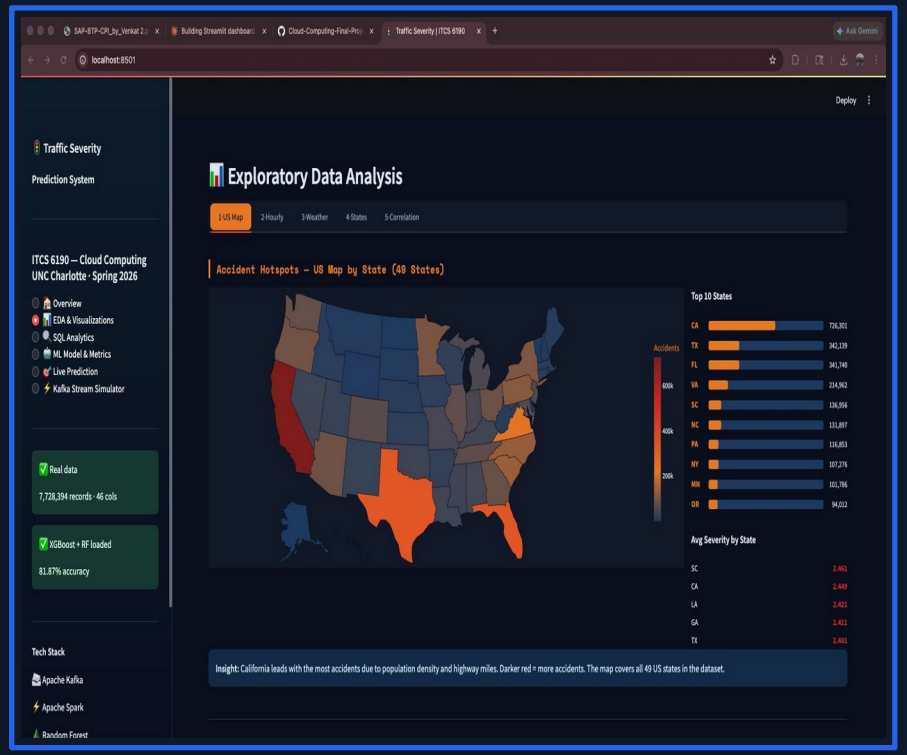
Streamlit Dashboard — Overview & EDA

🏠 Overview Page



7.7M records · 81.87% accuracy · Pipeline Architecture · Severity Scale

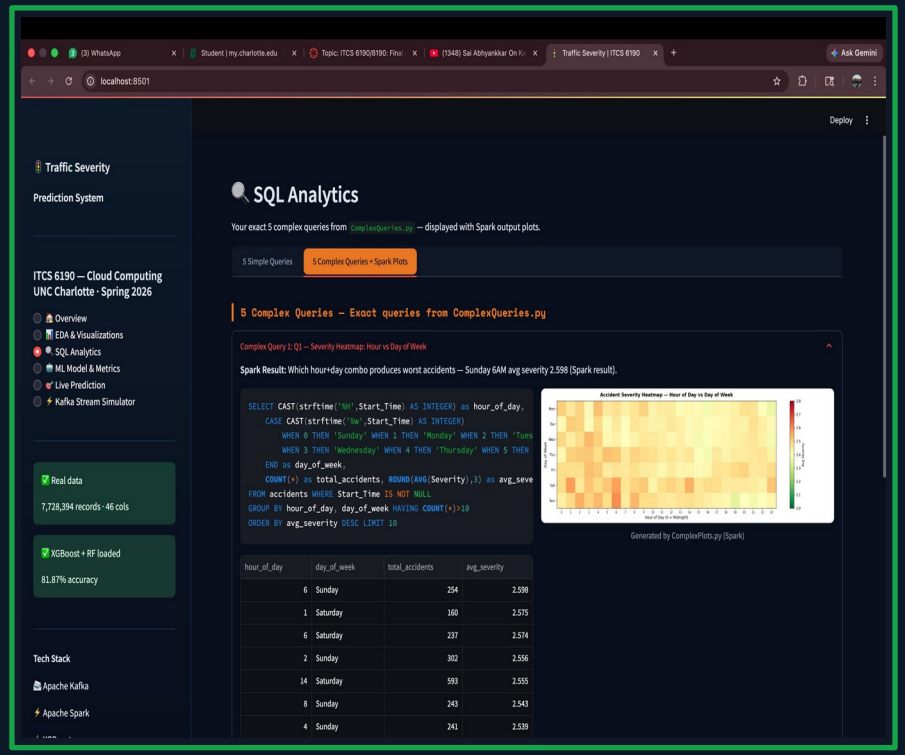
📊 EDA & Visualizations



US Map · Hourly Trend · Weather Risk · States · Feature Correlation

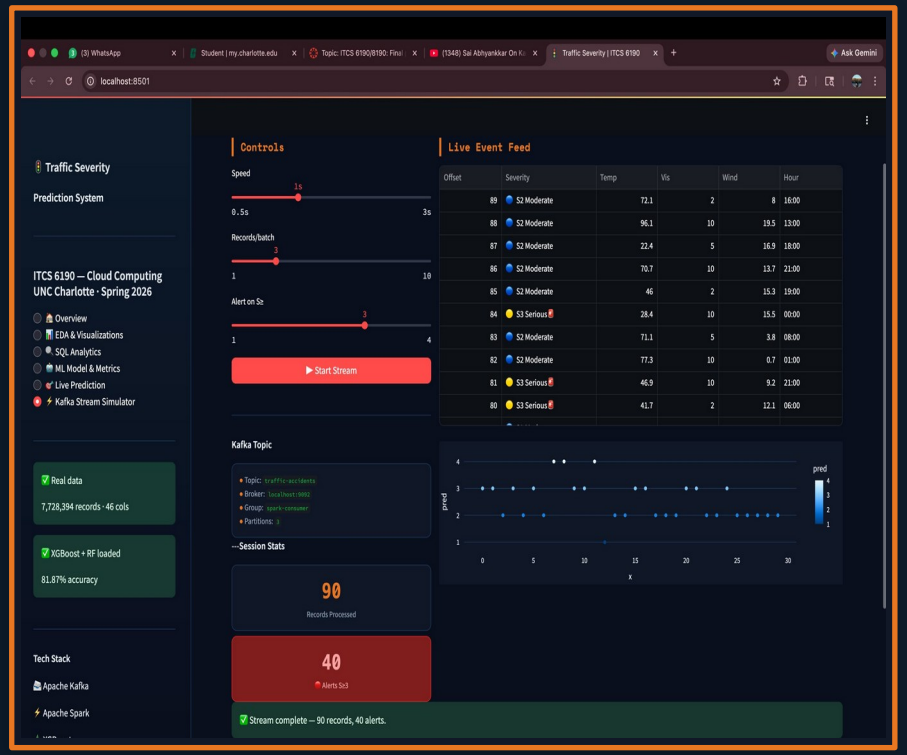
Streamlit Dashboard — SQL Analytics & Kafka Simulator

🔍 SQL Analytics



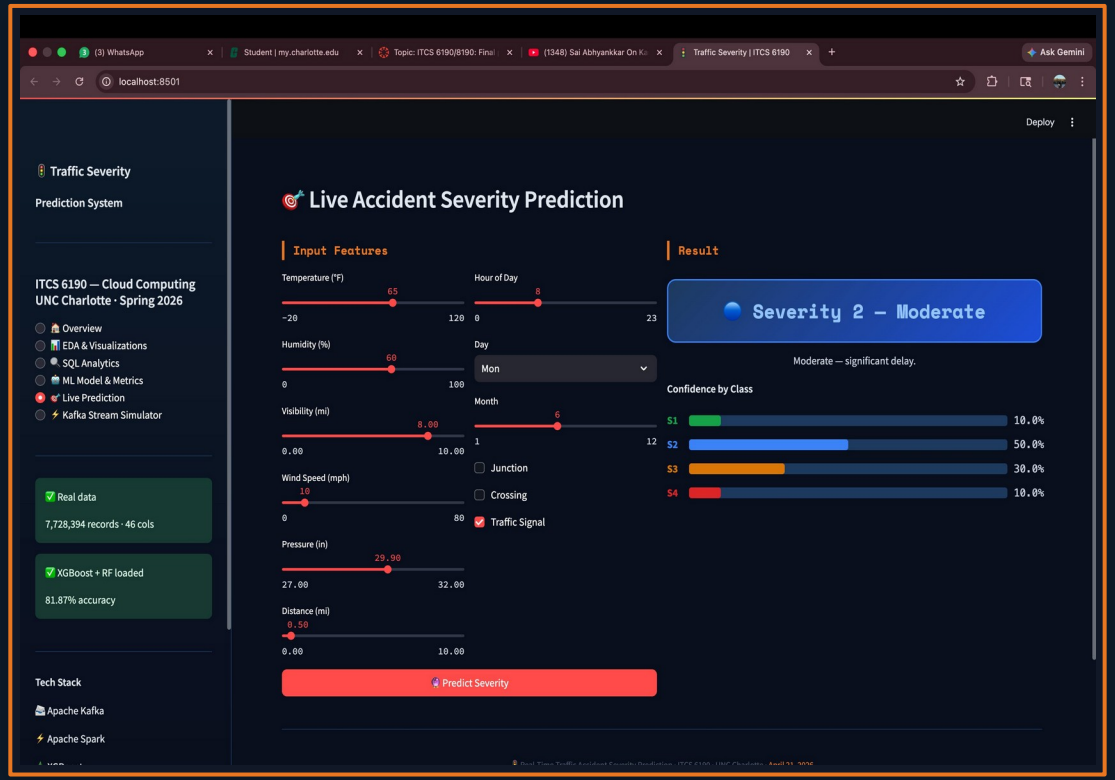
5 Complex Spark SQL Queries · Danger Score · Weather Risk · Road Segments

⚡ Kafka Stream Simulator

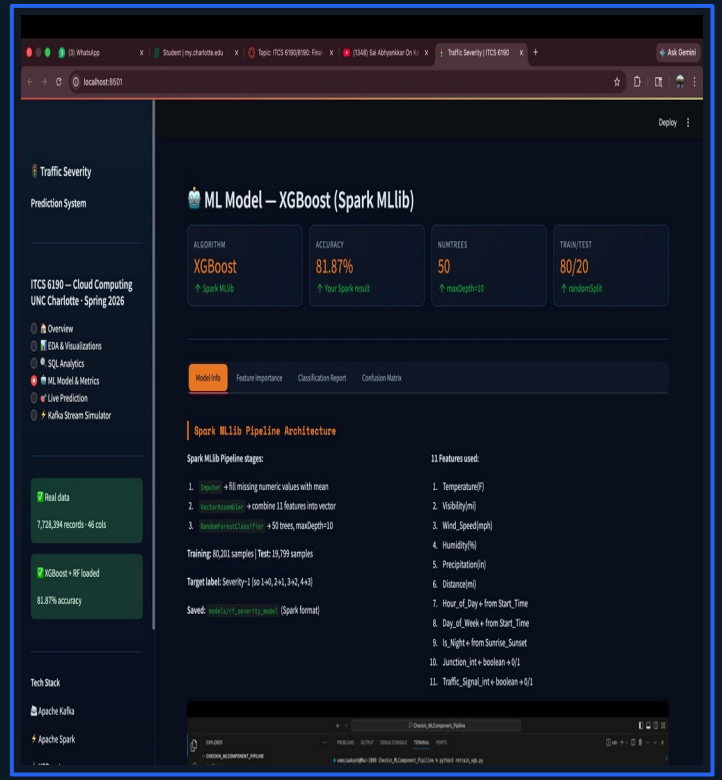


90 records processed · 40 alerts · Live severity stream · Real-time chart

Live Prediction & ML Model Dashboard



🎯 Live Prediction — S2 Moderate predicted



🤖 ML Model — 81.87% XGBoost

Real-time inference: input 15 features → XGBoost predicts Severity in milliseconds

Evaluation Metrics — How We Know It Works

Accuracy

Definition:

Out of all Severity predictions made, how many did the model get correct overall across all 4 classes.

Why we use it:

Quick overall performance snapshot. Easy to understand and communicate to non-technical stakeholders.

Note: Can be misleading — model always predicting Severity 2 still looks accurate due to class imbalance in dataset.

Weighted F1-Score

Definition:

Balances Precision and Recall for each Severity class individually, then combines weighted by class frequency.

Why we use it:

Primary metric. Handles class imbalance — penalises ignoring rare Severity 4 events. Far more honest than accuracy.

Note: Severity 4 is only 2.65% of data. Plain accuracy would hide the model completely failing on this critical class.

Confusion Matrix

Definition:

Table showing how many predictions landed in each Severity class — correct and incorrect — across all combinations.

Why we use it:

Reveals exactly where the model makes mistakes. Shows if Severity 4 is ever misclassified as a lower severity.

Note: Missing a Severity 4 prediction means suppressing an emergency alert. Most critical failure mode to check.

Future Upgrades — Next Steps

01 — Live Weather API

Connect real-time weather to predict dangerous conditions BEFORE any accident is reported. True prevention over reaction.

02 — Visual Map Dashboard

Live color-coded US map — green, yellow, red severity zones. Updates every 15 minutes. Usable by non-technical traffic managers.

03 — SMS and Email Alerts

Auto-send notifications when Severity 3 or 4 detected. Authorities warned instantly without manually checking output files.

04 — State-Level Partitioning

Separate Kafka topic per state. All 49 states processed in parallel. Eliminates bottlenecks during high-volume periods.

05 — Auto Model Retraining

Monthly scheduled retraining on new accident data. Old vs new model compared. Better model auto-deployed without manual intervention.

06 — Context Data Integration

Add public events, road construction, school schedules. System understands game-day vs normal — same road, completely different risk.

Conclusion

What We Built

A real-time big data pipeline streaming 7.73 million US accident records through Apache Kafka and Spark — detecting high-severity incidents, running live SQL hotspot analysis, and predicting Severity 1 through 4 using XGBoost (81.87% accuracy) trained on 15 weather, road, and time features — all within 15-minute windows.

Problem	Solution	Dataset	Model
Reactive systems warn drivers too late	Real-time streaming + ML severity prediction	7.73M records 46 cols · 49 States	XGBoost 81.87% + RF Spark 60.52%

This project shifts traffic accident management from reactive response to proactive intelligence.